

ON THE ASSESSMENT OF MUSCLE FATIGUE RATE VIA VARIOUS EMG FREQUENCY SPECTRAL PARAMETERS

Suebsak Nanthavanij

*Department of Mechanical and Industrial Engineering, New Jersey Institute of Technology,
Newark, NJ 07102 (U.S.A.)*

Subramaniam Deivanayagam

Department of Industrial Engineering, Tennessee Technological University, Cookeville, TN 38505 (U.S.A.)

(Received January 27, 1988; accepted in revised form February 19, 1989)

ABSTRACT

A research study was undertaken to investigate the fatigue process of quadriceps muscle during sustained isometric contraction at several work load levels using root-mean-squared (RMS) voltage and frequency parameters of EMG frequency spectral distribution. Its objective was to evaluate the possibility of using those parameters to indicate muscle fatigue rate. Twelve male subjects participated in an experiment which required them to exert and maintain isometric knee extension forces at 25%, 40%, 55%, and 70% of their maximal voluntary contraction (MVC) force levels. The results showed that both initial mean power frequency (MPF) and median power frequency (MF) (as determined from MPF and MF versus time curves) were not significantly influenced by muscle contraction level ($p > 0.1$). RMS voltage, MPF, and MF linear slopes, on the other hand, were significantly correlated to muscle contraction level ($p < 0.001$) which suggested that they could be used to represent muscle fatigue rate. The exponential relationship found in this study showed that quadriceps fatigued at an increasing rate as the work load increased. Further, an improvement in the coefficient of correlation between their linear slopes and muscle work load was found when expressing muscle work load in terms of the ratio of force to body weight. This finding indicated the possibility of using body weight as a basis of determining work load level which could help to improve the quality of quantitative assessment of muscle fatigue.

INTRODUCTION

For decades electromyogram (EMG) has been used to study the contraction of a muscle and its fatigue characteristics. Briefly, EMG signals are the complex electrical signals resulting from an integrated firing of active motor units in a working muscle. These signals can be picked up by

either implanted or surface electrodes. They are usually interpreted in terms of integrated EMG voltage and certain frequency spectral distribution parameters such as mean power frequency and median power frequency (MPF and MF, respectively).

A linear relationship between integrated EMG voltage and muscle tension was reported by several

researchers (Inman et al., 1952; Edwards and Lippold, 1956; DeVries, 1968a; Milner-Brown and Stein, 1975; Lind and Petrofsky, 1979; Pollak, 1980). If a muscle has to maintain a certain level of tension over a prolonged period, it gradually fatigues. The number of active motor units and their firing frequencies will be increased, resulting in an increase in the amplitude of the integrated EMG signal (Edwards and Lippold, 1956; Eason, 1960; DeVries, 1968b; Lloyd, 1971).

Studying the carrier wave of surface EMG during a fatiguing exercise, Piper (1912) found a reduction in the frequency of its spectrum. Since then, the changes in frequency spectral pattern of EMG signals have been extensively studied using a technique called frequency spectral analysis. Its spectral content was found to shift towards the lower frequencies during a sustained muscle contraction (Kogi and Hakamada, 1962; Kaiser and Petersen, 1963; Kadehors et al., 1968; Lloyd, 1971; Duxbury et al., 1976; Lindstrom et al., 1977; Viitasalo and Komi, 1977; Petrofsky, 1980; Hagberg, 1981; Naeije and Zorn, 1981, 1982; Moritani et al., 1986). This spectral shift could be detected by a decrease in either MPF or MF of the EMG frequency spectral distribution density.

Surprisingly, the relationship between EMG frequency spectrum and muscle tension is still inconclusive. During brief static exercises, Viitasalo and Komi (1978b), Petrofsky (1980), Petrofsky and Lind (1980), Bigland-Ritchie et al. (1981), and Merletti et al. (1984) did not find any significant changes in the frequency spectral distribution of EMG (as indicated by either MPF or MF) when the muscle tension was changed. Guha and Anand (1979), however, derived a simulation model of motor unit recruitment and proposed that when muscle tension increased, the high-to-low ratio of EMG spectral power would increase. Hagberg and Ericson (1982) found that MPF increased with muscle contraction at low levels. At the levels above 25–30% of MVC, MPF became independent of contraction level. MF was also reported to have a curvilinear relationship with contraction level by Van Boxtel and Schomaker (1984). For fatiguing static exercises, a greater decrease rate of MPF and MF at higher muscle tensions was reported (Viitasalo and Komi, 1978a; Naeije and Zorn, 1981; Sadoyama et al., 1983). Petrofsky and Lind (1980), Sadoyama and Miyano

(1981), on the other hand, did not find any significant difference between MPF at any fraction of time duration between different muscle tensions.

In summary, changes in EMG frequency spectral distribution density were found when a muscle underwent strenuous exercise which caused it to fatigue. These changes could be detected by decreases in MPF and MF, and an increase in RMS voltage. During brief, nonfatiguing exercises at different contraction levels, several researchers concluded that there were no significant changes in the frequency spectral distribution of EMG while some found different results. For fatiguing static exercises, non-uniform results were also reported.

This research study was conducted to investigate the fatigue process of quadriceps muscle while maintaining a constant isometric force application effort at different levels. Since it has been known that a muscle fatigues at a faster rate at higher muscle work loads, the objective of this study was to determine if any of the EMG parameters could be used to indicate this fact and provide a means to represent muscle fatigue rate. Quadriceps muscle was selected due to its accessibility, its role in supporting and maintaining body posture, and its involvement in many physical activities. Due to the inconclusive results from previous findings regarding the relationships between the EMG parameters and muscle contraction, several parameters were considered. They were initial MPF, initial MF, RMS voltage linear slope, MPF linear slope, and MF linear slope of the EMG frequency spectral distribution density.

METHODS

Twelve healthy male subjects with no known history of knee injury voluntarily participated in this study. All subjects were explained the purpose of this study and the experimental procedures. They then signed an informed consent form to participate. The subjects' ages, body weights, and body heights ranged between 21–34 years ($\bar{X}$ = 26), 53–76 kg ($\bar{X}$ = 63), and 1.63–1.80 m ($\bar{X}$ = 1.72), respectively.

The subject was seated on a specially built chair in an upright posture with his legs flexing naturally (right angle) at the knee joints. Both legs

were kept behind a padded leg bar which was connected to a strain gauge load cell. In order to avoid any knee movement during the experiment, a padded knee frame was placed over both thighs and adjusted so that it did not stop the blood flow and still provided enough resistance for knee movement. Seat belts were also used to restrain body movement. The leg bar was adjusted so that it was at the same level as the anterior surface of the distal end of subject's tibia, just above his ankle.

The experiment was divided into two parts, i.e., a maximal voluntary contraction (MVC) force measurement and a fatigue study.

MVC force measurement

The subject was asked to isometrically push his right lower leg forward against the leg bar as hard as he could and maintain that maximal effort for five seconds. During this activity, the subject's right lower leg was maintained at 90 deg from his thigh; thus, the resulting MVC force represented his maximal knee extension force at 90 deg angle. The MVC force measurement was performed three times with five minute rest periods allowed in between two successive trials. Peak MVC forces were monitored by an experimenter via a digital strain indicator which was connected to the strain gauge load cell and recorded on a tape by a FM tape recorder. Then the average MVC was calculated from the two closest peak values.

Fatigue study

Four submaximal levels, namely, 25%, 40%, 55%, and 70% of MVC force, were calculated and then applied in random order to study the fatigue process of quadriceps. At each fatigue trial, a force feedback oscilloscope was set such that a vertical movement of a lighted dot (shown on the screen) represented the changes in the subject's knee extension force level. Additionally, when the force reached a predetermined level, the lighted dot would be at the center of the oscilloscope screen. This oscilloscope was placed in front of the subject and served as a feedback of his effort in maintaining a constant level of knee extension. The subject was then instructed to gradually generate his knee extension force and maintain a constant effort by keeping the lighted dot at the center of the screen. The 25% and 40% of MVC fatigue trials lasted 40 s while the other two fatigue trials lasted 30 s. Seven minute rest periods were allowed between two consecutive fatigue trials for the muscle to recover.

EMG signals were monitored by three sets of silver-silver chloride surface electrodes (4 mm diameter) placed over vastus lateralis, rectus femoris, and vastus medialis muscles of the right thigh. The skin was carefully prepared according to the standard procedures and the electrodes were placed longitudinally over the belly of individual quadriceps muscle components. Electrode locations were also recorded on the data sheet. EMG

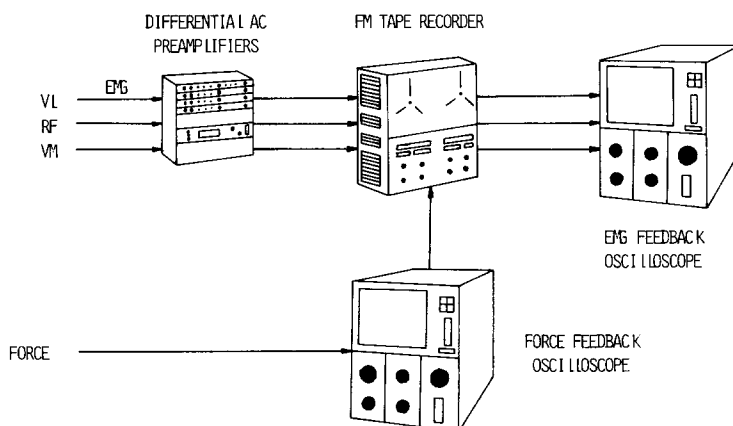


Fig. 1. A flow chart showing a set-up of the data recording instruments.

signals were amplified with a differential amplifier (Grass Instruments, P511K) and stored on a tape (recorder: Hewlett Packard, 3968A). A set-up of the data recording instruments is shown in Fig. 1.

The EMG feedback oscilloscope (see Fig. 1) was set up so that only the experimenter could monitor the EMG signals from quadriceps muscle components. Its purpose was to enable the experimenter to observe the quality of the EMG signals being collected. During the fatigue study, the recording of EMG signals was started when the subject was able to generate his knee extension force up to the predetermined level.

The entire experiment was performed three times with at least one week separation between the two test days. The experimental procedures were the same for every test but another random order of the submaximal exercises was followed. At each time, the subject's MVC force was remeasured and used to calculate the four submaximal contraction levels for fatigue trials. Electrodes were also placed at the same locations to assure the reliability of the recorded EMG signals.

DATA ANALYSIS

A frequency spectral analysis of EMG signals was performed by a DEC LSI-11/23, 16-bit minicomputer. The recorded signals were reproduced by the FM tape recorder and filtered with a four-pole low-pass Butterworth filter set at 600 Hz and a four-pole high-pass Butterworth filter set at 2 Hz. They were sampled at a rate of 2,048 Hz by a 12-bit analog-to-digital converter. The digital data was further processed in 250 millisecond segments by multiplying by a trapezoidal window function and transforming by a 512-point complex fast Fourier transform (FFT) to produce a frequency spectral periodogram. Then four consecutive periodograms were averaged and used to calculate the RMS voltage and frequency spectral parameters for the frequency distribution density ranging from 12 to 512 Hz of each second. The RMS voltage, MPF, and MF of EMG frequency spectral distribution density were plotted against time. A regression analysis was then performed to determine the y -intercepts and slopes of their linear relationships.

Additionally, muscle forces recorded during the

fatigue trials were also digitized at a sampling rate of 2 Hz. The average force was then found by summing all digitized values and dividing by the total number of sample points.

The averages of individual parameters of EMG frequency spectral distribution density from the three quadriceps muscle components were calculated and used as representatives of muscle activity. In order to determine if any of them could be used to indicate the muscle fatigue rate, initial MPF, initial MF, RMS voltage linear slope, MPF linear slope, and MF linear slope were tested to determine if they were significantly influenced by different muscle contraction levels and different test days using two-factor analysis of variance. Initial frequency spectral parameter was considered as the y -intercept of a plot of frequency parameter against time. Since both MPF and MF decreased with time and only the rates of their decrease (not their direction) were the main interest in this study, the absolute values of their linear slopes were used. The term 'linear slope' appearing thereafter in this article thus represents the absolute value. Finally, a regression analysis was performed to explain the relationship between muscle fatigue rate and muscle contraction.

RESULTS

Initial frequency parameters

Table 1 shows the averages of initial MPF and MF for four muscle contraction levels and three test days. The data shows slight increases in both initial MPF and initial MF as muscle contraction

TABLE 1

The averages of initial MPF and MF of each % of MVC and test day ^a

Initial frequency	Test day	% of MVC			
		25%	40%	55%	70%
MPF	1	75.55	77.90	78.14	78.69
	2	76.97	77.62	78.95	78.57
	3	73.44	75.46	77.15	78.20
MF	1	62.59	63.58	63.89	64.75
	2	63.51	64.37	65.98	65.30
	3	59.99	62.47	63.70	64.85

^a The unit of initial MPF and MF is Hz, and each value represents the average of 12 subjects.

level increases. However, using two-factor analysis of variance, the differences between the initial MPFs corresponding to the four levels of submaximal contraction were not significant ($p > 0.1$). When comparing the initial MPFs of each contraction level for the three test days, the differences were found to be insignificant ($p > 0.1$). Initial MFs also showed similar behavior in both comparisons. No significant differences were found ($p > 0.1$).

RMS voltage linear slope

RMS voltage linear slope was calculated from a plot of RMS voltage against time. The data is presented in Table 2. A statistical analysis showed that the differences between RMS voltage linear slopes of four contraction levels were significant ($p < 0.001$). Nevertheless, its values seemed to show good consistency on test-to-test basis since no significant differences were found when compared for the three test days ($p > 0.5$).

MPF and MF linear slopes

Figure 2 shows a typical plot of MPF against time of one subject. Each line represents the MPF of EMG frequency spectral distribution density from each quadriceps muscle component. It can be seen that the MPF linearly decreases with time and its linear slope becomes steeper as muscle contraction level increases. Table 2 also confirms

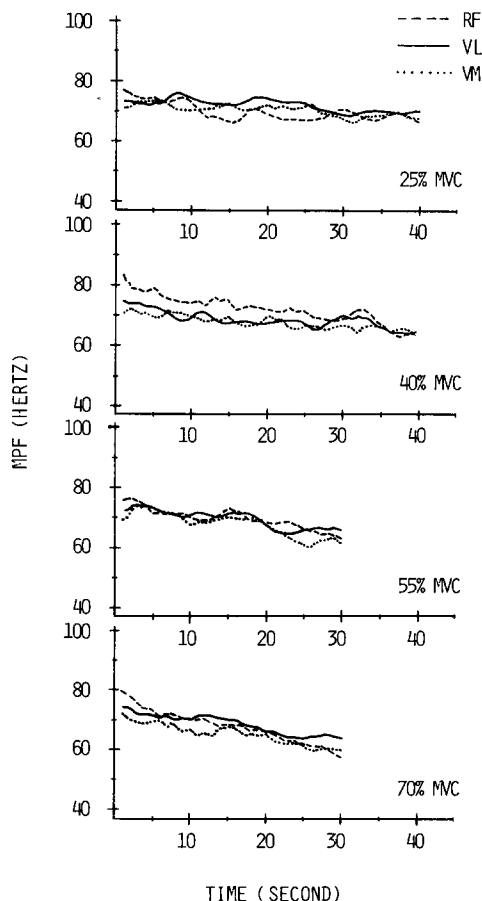


Fig. 2. Plots of MPF against time of three quadriceps muscle components [vastus lateralis (VL), rectus femoris (RF), and vastus medialis (VM)] during sustained isometric knee extension.

TABLE 2

Linear slopes ^a of RMS voltage, MPF, and MF

Linear slope	Test day	% of MVC			
		25%	40%	55%	70%
RMS voltage ^b	1	0.64	1.13	2.25	3.25
	2	0.63	1.02	2.11	3.21
	3	0.64	1.28	1.81	2.24
MPF ^c	1	0.18	0.23	0.40	0.58
	2	0.16	0.23	0.40	0.51
	3	0.13	0.20	0.36	0.54
MF ^c	1	0.16	0.19	0.34	0.47
	2	0.14	0.22	0.41	0.44
	3	0.13	0.19	0.31	0.48

^a Each value represents the average of 12 subjects.

^b The unit of RMS voltage linear slopes is microvolts/s.

^c The unit of MPF and MF linear slopes is Hz/s.

the differences of MPF linear slopes between all four contraction levels. These differences were found to be statistically significant ($p < 0.001$). A similar result was also found in the case of MF linear slope (also shown in Table 2). Comparing the three test days, the differences between either MPF or MF linear slope of the same contraction level were not significant ($p > 0.5$).

Since only three parameters, namely, RMS voltage linear slope, MPF linear slope, and MF linear slope were significantly influenced by muscle contraction, each was then plotted against muscle contraction (% of MVC), as shown in Fig. 3a–c. All linear slopes show a positive relationship with muscle contraction. Since it is known that rate of muscle fatigue increases with the level of contraction, the increases in those linear slopes with

muscle contraction therefore implicitly represent muscle fatigue rate. From Fig. 3a–c, an increase in each parameter is shown to be more pronounced at higher work loads. In other words, the relationship between each parameter and muscle contraction could be explained by an exponential model. The three models which express the relationships between various muscle fatigue rate indicators and muscle contraction are given below.

RMS voltage linear slope

$$= [0.2551] \times [1.0323]^{\% \text{ of MVC}} \quad (r = 0.62) \quad (1)$$

MPF linear slope

$$= [0.0686] \times [1.0286]^{\% \text{ of MVC}} \quad (r = 0.72) \quad (2)$$

MF linear slope

$$= [0.0610] \times [1.0287]^{\% \text{ of MVC}} \quad (r = 0.70) \quad (3)$$

These three models suggest that the rate of muscle fatigue increases exponentially with muscle contraction for the range of contraction levels studied. The dispersion of each parameter was noted in all four contraction levels, but was more pronounced at higher levels (see Fig. 3). This wide dispersion may be attributed to the inter-personnel differences always seen in any experiment involving the use of human subjects. One possible explanation was that individual subjects had different muscle fiber compositions which might influence both the rate of fatigue and EMG signals as well.

In an attempt to find a better way of correlating fatigue rate with muscle contraction, another muscle work load normalizing factor was considered, i.e., subject's body weight. Initially, it was observed that some subjects with lower body weight were able to generate MVC forces comparable to the subjects with higher body weight. In addition, their rates of fatigue were consistently higher than the others in every muscle contraction level. Therefore, the ratio of force to body weight (F/BW) was used as another muscle work load expression, in place of percentage of MVC. The plots of muscle fatigue rate indicators against the ratio of force to body weight are shown in Fig. 4a–c and their exponential relationships are given below.

RMS voltage linear slope

$$= [0.2461] \times [87.7737]^{F/BW} \quad (r = 0.72) \quad (4)$$

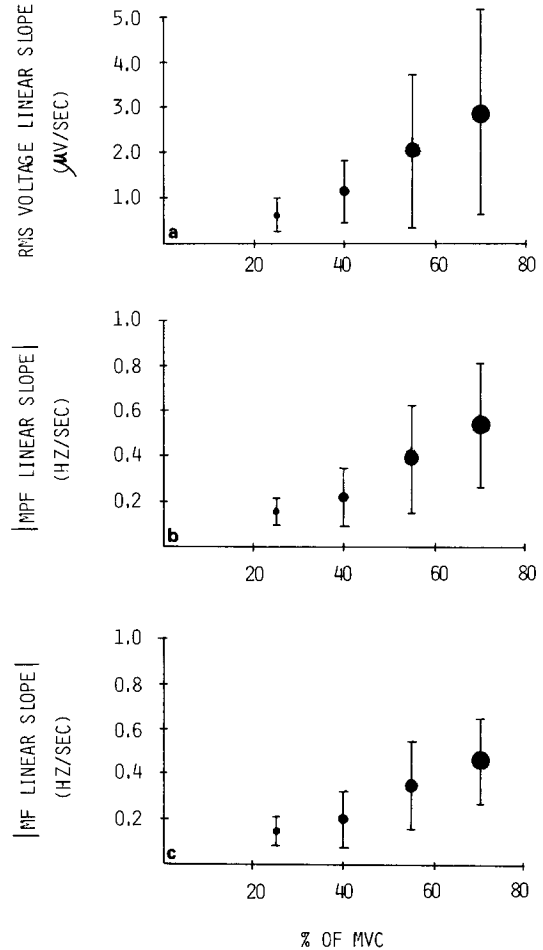


Fig. 3. Plots of EMG frequency spectral parameters against muscle contraction (expressed in terms of a percentage of MVC): (a) RMS voltage linear slope, (b) MPF linear slope, and (c) MF linear slope.

MPF linear slope

$$= [0.0732] \times [39.7481]^{F/BW} \quad (r = 0.78) \quad (5)$$

MF linear slope

$$= [0.0677] \times [36.9082]^{F/BW} \quad (r = 0.77) \quad (6)$$

The correlation coefficients of the models using the ratio of force to body weight were also consistently higher than the models which used percentage of MVC for every muscle fatigue rate indicator, as presented in eqns. (1)–(6).

The results obtained from this research study therefore demonstrated the possibility of using RMS voltage, MPF, and MF linear slopes as representatives for muscle fatigue rate and pre-

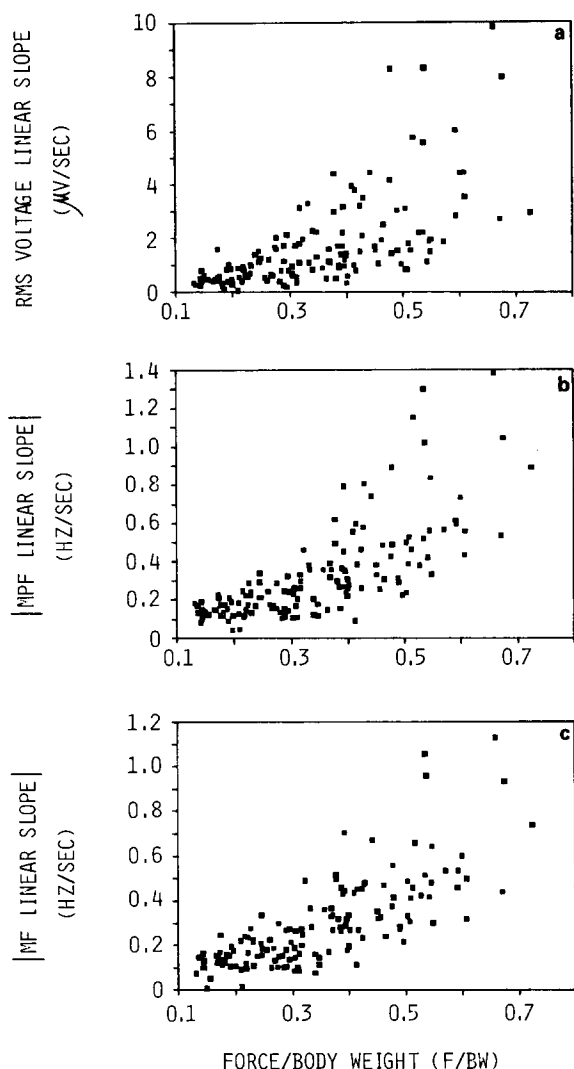


Fig. 4. Plots of EMG frequency spectral parameters against muscle contraction (expressed in terms of the ratio of force to body weight): (a) RMS voltage linear slope, (b) MPF linear slope, and (c) MF linear slope.

sented an exponential relationship between rate of fatigue and muscle contraction.

DISCUSSION

Localized muscle fatigue is generally defined as an inability of the muscle to maintain the required expected muscle output (Grandjean, 1981; Astrand and Rodahl, 1986). Muscle fatigue is a complex phenomenon since it involves both psychological and physiological factors. Discomfort sensation

and pain are usually found to accompany muscle fatigue induced by an isometric type of contraction. It seems to appear that this definition applies only to any fatigue study which requires the subject to maintain his/her constant muscle output at a certain level until it decreases to an unacceptable level. This implies that (the onset of) muscle fatigue occurs when the degradation of performance is found to be unacceptable. In this quadriceps fatigue study, the subjects were required to maintain their muscle output for a certain time duration. No degradation of performance was found; yet, their quadriceps did fatigue. It becomes clear that the term 'muscle fatigue' used in this study must be redefined.

Changes in the EMG frequency spectral parameters were found in several fatigue studies and were widely used as the parameters to investigate muscle fatigue characteristics. Signal changes first occurred at the beginning and continued until the cessation of muscle activity. The knowledge gained from those findings thus far enables us to redefine muscle fatigue accordingly.

Muscle fatigue can be defined as a combined psychological and physiological phenomenon which imposes upon individuals as a result of an effort to maintain a required expected muscle output. Muscle fatigue occurs at the initiation of the required muscle activity, increases along the time course, and eventually causes the muscle to fail to maintain its required output. The process of muscle fatigue depends on several factors, such as individual characteristics, type of activity, the required level of muscle output, etc. According to the definition given above, our analysis of the muscle fatigue process considers the onset of fatigue as the time moment where the muscle output first reached the required level. Any effect resulting from 'muscle warm-up or recruitment' on the EMG signals collected was therefore excluded. The experimental procedures were designed so that this effect was minimized. These included allowing the subject to practice his force application only once or twice before starting data collection, providing rest period between the practice and the actual activity, and instructing the subject to 'gradually push' and 'try to reach the required level within three seconds'.

Since the entire experiment was repeated three times and four levels of muscle activity were per-

formed each time, it was anticipated that the results obtained from the subsequent fatigue trials and test days could be affected by a training effect. Attempts were made to minimize this effect by assigning the order of the four fatigue trials randomly, changing the random order every test day, and providing at least one week separation between two test days.

An isometric knee extension type of exercise is one of several muscle activities which require the cooperation of several muscle components to carry out an assigned task. In general, knee extension is mainly performed by the contraction of vastus lateralis, vastus intermedius, vastus medialis, and rectus femoris, together known as quadriceps. The function of either rectus femoris (Viitasalo and Komi, 1975, 1977) or vastus lateralis (Thorstensson and Karlsson, 1976; Komi and Tesch, 1979) was individually investigated in the studies involving this type of exercise. It was suspected that their results could have lost some accuracy since the behavior of only one quadriceps muscle component was used to explain the behavior of the whole group. In this study, three muscle components were investigated. They were rectus femoris, vastus lateralis, and vastus medialis. Vastus intermedius was excluded since it was not possible to monitor its activity with surface electrodes.

During data analysis, it was noticed that while maintaining a constant force application, quadriceps occasionally shifted its activity among its muscle components as detected by a sudden decrease of the frequency spectral parameter in one muscle while a simultaneous change in the opposite direction was found in the other muscle(s), as shown in Fig. 5. This shift of load between muscle components occurred when one component became 'relaxed', causing another component to 'increase' the level of its activity in order to maintain a constant muscle power output. It is suspected that this phenomenon may influence the MPF and MF of individual quadriceps muscle components and, subsequently, their linear slopes. Therefore, in order to normalize this possible effect, EMG frequency spectral parameters (initial frequencies and linear slopes) from all three muscle components were averaged and used as representatives of muscle activity (Nanthavanij, 1987).

Among the five parameters considered, none was found to be significantly influenced by differ-

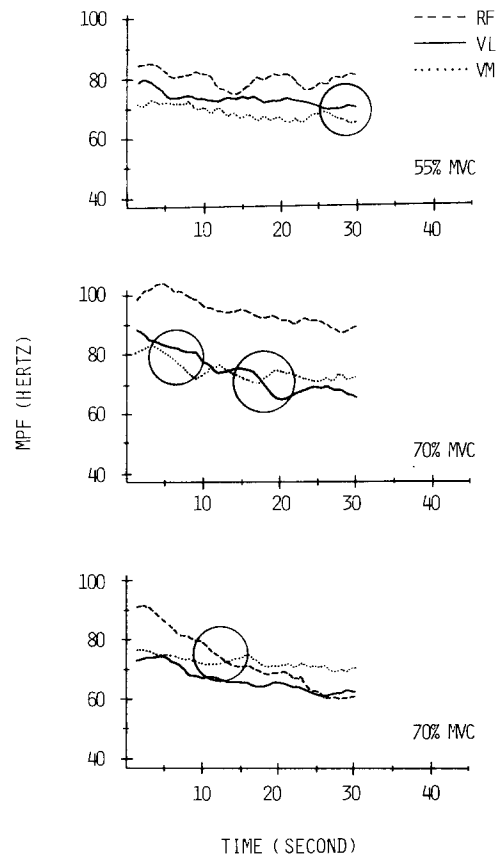


Fig. 5. A shift of activity among quadriceps muscle components.

ent test days. Similar results were also reported by Komi and Buskirk (1970), Viitasalo and Komi (1975), and Duxbury et al. (1976), despite different muscles and types of exercise studied. This consistency of EMG signals reflected the ability of a muscle to reproduce electrical signals of similar characteristics under essentially the same work load and other conditions. The dispersion of individual parameters at every contraction level was clearly seen and could have resulted from the use of MVC force to normalize inter-personnel differences. Since MVC forces were remeasured and subsequently used in calculating four submaximal contraction levels, quadriceps muscle components actually worked at different levels of force on different test days. Even though this variation did not cause significant changes in EMG parameters when compared between different test days, it was suspected that its effect on their correlations to muscle contraction could be significant. This fact

could have been overlooked if the contraction level was expressed in terms of a percentage of MVC. In addition, differences in muscle fiber composition may influence the EMG signal characteristics as well. Viitasalo and Komi (1978a), however, did not find any significant changes in RMS voltage at different muscle fiber composition. Therefore the wide dispersion of RMS voltage linear slope may be mainly attributed to the effects of using MVC force as a normalizing factor.

Statistical comparisons between the four contraction levels found both initial MPF and MF to be unchanged. These results were similar to those reported by Viitasalo and Komi (1978b), Petrofsky and Lind (1980). Their values were in the ranges of 73–78 Hz for initial MPF, and 60–66 Hz for initial MF. Their constancies suggested that EMG signals recorded at the beginning of any sustained muscle contraction had the same frequency content distribution irrespective of the contraction level. One possible explanation is that in order to generate muscle forces of different levels, different numbers of motor units, both slow twitch (ST) and fast twitch (FT), are recruited depending on the contraction level. However, the ratio of the number of ST units to FT units recruited may stay relatively constant, resulting in a similar distribution pattern of its frequency spectrum. Therefore, no significant changes in the parameters derived from the EMG frequency content distribution are found among different contraction levels.

While a muscle was maintaining a constant force application, decreases in MPF and MF, along with an increase in RMS voltage were found, and could be related to the onset of muscle fatigue (similar to the results found in several studies). The relationships between these three parameters and time course were found to be linear at all muscle contraction levels studied. Since the changes in these parameters and the onset of muscle fatigue occurred simultaneously, it was reasonable to believe that rates of changes in these parameters (their linear slopes) were also related to rates of muscle fatigue. In order to test whether these linear slopes could be used to indicate fatigue rate, each was plotted against muscle contraction level. The results from statistical analysis showed that the parameters' linear slopes changed

with muscle contraction and could be satisfactorily explained by an exponential model. It was then concluded that the linear slopes of MPF, MF, and RMS voltage were good indicators for muscle fatigue rate.

Chaffin (1969) also reported an accelerating increase in muscle fatigue rate as the load increased. In his study, the muscle under investigation was the biceps brachii and the EMG frequency spectral parameter used was the % of total EMG power in two different frequency bands, namely, 4–30 Hz and 60–100 Hz. Though being unable to numerically compare the results obtained from this study with his findings due to the differences in muscle groups, the parameters used, and the expressions of muscle work load, it can be stated that the fatigue process of quadriceps is similar to that of biceps brachii.

By visually inspecting a plot of linear slope against muscle contraction (see Fig. 3), it was noted that the dispersion from a mean value was more pronounced at higher contraction levels. It was first suspected that this dispersion was caused by the fluctuation of the subjects' knee extension force application while trying to maintain a constant level. If any subject's force output fluctuated highly, its mean value could deviate significantly from the predetermined level and could have caused a wide dispersion of EMG frequency spectral parameters when being plotted against % of MVC. Controlling a constant muscle performance at a higher output level was expected to be more difficult than that at a lower level, causing a wider dispersion as the force level increased. However, an analysis of the digitized force values along the time course showed good constancy at every force level, and in every subject. Thus, the dispersion found in this study was not possibly due to the force fluctuation. Viitasalo and Komi (1978a) explained that this phenomenon was due to the difference in muscle fiber composition found between subjects. A subject with his muscle fiber composition predominated by FT fiber type was more susceptible to muscle fatigue than the one with his muscle fiber composition predominated by ST type, under the same work load (Komi and Tesch, 1979). This could result in a higher muscle fatigue rate usually found in a FT fiber predominant subject than that in a ST fiber predominant one. Since at lower contraction levels most of the

FT motor units remained inactive, the effect of muscle fibers composition on muscle fatigue was expected to be less and could be seen by narrower dispersion of the EMG frequency spectral parameters. The results of this study, therefore, agreed with the previous findings.

However, it must be mentioned that muscle work load is usually expressed in terms of a percentage of MVC force in order to reduce the inter-personnel differences. Its reliability was doubtful to some researchers (DeVries, 1968a; Kondraske et al., 1987). It was noted in this study that a subject with lower body weight who was able to generate his maximum force comparable to the subject with higher body weight had a consistently higher muscle fatigue rate at every work load. This phenomenon could be partially responsible for a wide dispersion found when plotting each parameter against muscle contraction expressed in terms of a percentage of MVC. Kondraske et al. (1987) used body weight to normalize this dispersion in their back muscles study and found that the ratio of force to body weight showed a better correlation with a fatigue rate indicator. They hypothesized that one of the major functions of back muscles was to support and maintain body posture. Therefore it was anticipated that their capability was more closely related to body weight. This research study also found a similar result when substituting MVC force by body weight. This similarity was expected since quadriceps also has to support body weight as its routine function. It was found, as shown in Fig. 4, that for the subjects with lower body weights, using the ratio of force to body weight as x -axis caused the EMG frequency spectral parameters to move to the right. A reverse direction of movement of those parameters was also found for the subjects with higher body weights. An improvement in a correlation coefficient was found with a new muscle work load expression as a result of this partial rotation, but the importance of differences in muscle fiber composition could not be ignored. It was mentioned in the literature that motivation could positively or negatively influence the results obtained from the MVC force measurement (Kroemer, 1974). It was also suspected in this study that the effect of motivation on the MVC force was one of the factors which caused the dispersion of EMG frequency spectral

parameters found in Fig. 3. The use of body weight as a normalizing factor in this fatigue analysis was able to partially eliminate the effect of some of the factors thought responsible for such dispersion. Though body weight seems to be a promising factor to normalize the inter-personnel differences, extensive studies especially those involving different muscle groups, subjects with different levels of physical fitness, etc. are required to confirm its usefulness in muscle fatigue studies.

In conclusion, EMG frequency spectrum of quadriceps at the onset of the fatigue has relatively the same frequency content distribution for every muscle contraction studied as indicated by the constancy of initial MPF and MF. During sustained contraction, quadriceps fatigues at a constant rate throughout the fatigue trial. Its fatigue rate exponentially correlates with the level of contraction over the range studied. RMS voltage, MPF, and MF linear slopes were found to be good indicators of muscle fatigue rate since they were not significantly influenced by different test days. Body weight also shows strong potential of being a better normalizing factor to reduce the inter-personnel differences, especially for postural muscles such as leg muscles and back muscles.

The significance of this work is the potential effect it can have on muscle rehabilitation programs. While industrial engineers, bioengineers, industrial safety engineers, physiologists, physical therapists, and ergonomists have designed many studies to assess human lifting limits, strength, and work capacity, only limited clinical applications have resulted. Thus far, periodical measurements of muscle strength during rehabilitation period have been used as a tool to determine the relative improvement of muscle function. Quantitative assessment of muscle fatigue may prove to be more fruitful than the technique described above since it can be done under submaximal contraction of rehabilitating muscle, reducing pain and overexhaustion usually experienced by individual by limiting exercise duration, yet providing informative results pertaining muscle functional status. Further, the use of body weight as a basis for determining load level helps to decrease some effects on assessment of physiologic function from submotivated or malingering subjects.

In an industrial environment, the results of this

work can be applied to the tasks which require the workers' muscular effort of a certain level. Though a test of MVC force proves to provide useful data for a personnel screening process, the quantitative knowledge of muscle fatigue characteristics is believed to supply more realistic and applicable information regarding human physical performance on a prolonged activity. It should be noted, however, that a norm of acceptable muscle fatigue rate must be established based upon the type of task involved, active muscle groups, and the workers' physical characteristics. The benefits gained from using this rate of muscle fatigue in personnel screening may help to reduce, in some degree, the number of workers suffering from muscle injuries and thus improve the safety in work environment.

ACKNOWLEDGEMENTS

The authors wish to acknowledge Timothy W. Carmichael, Robert W. Butsch, and Dr. George V. Kondraske for their invaluable suggestions and assistance. Equipment and computer facilities were provided by the Veterans Administration Medical Center at Dallas. This research study was partially funded by the Veterans Administration Medical Center, Grant No. 822, Dr. Vert Mooney, Principal Investigator.

REFERENCES

- Astrand, P. and Rodahl, K., 1986. *Textbook of Work Physiology: Physiological Bases of Exercise*. McGraw-Hill, 3rd edn, 756 pp.
- Bigland-Ritchie, G., Donovan, E.F. and Roussos, C.S., 1981. Conduction velocity and EMG power spectrum changes in fatigue of sustained maximal efforts. *J. Appl. Physiol.*, 51: 1300-1305.
- Chaffin, D.B., 1969. Electromyography—A method of measuring local muscle fatigue. *J. Methods-Time Meas.*, XIV: 29-36.
- DeVries, H.A., 1968a. Efficiency of electrical activity as a physiological measure of the functional state of muscle tissue. *Amer. J. Phys. Med.*, 47: 10-22.
- DeVries, H.A., 1968b. Method for evaluation of muscle fatigue and endurance from electromyographic fatigue curves. *Amer. J. Phys. Med.*, 47: 125-135.
- Duxbury, A.J., Hughes, D.F. and Clark, D.E., 1976. Power spectral distribution of the masseter electromyogram from surface electrodes. *J. Oral Rehabil.*, 3: 333-339.
- Eason, R.G., 1960. Electromyographic study of local and generalized muscular impairment. *J. Appl. Physiol.*, 15: 479-482.
- Edwards, R.J. and Lippold, O.C.J., 1956. The relation between force and the integrated electrical activity in fatigued muscles. *J. Physiol.*, 132: 677-681.
- Grandjean, E., 1981. *Fitting the Task to the Man: An Ergonomic Approach*. Taylor & Francis Ltd., 3rd edn.
- Guha, S.K. and Anand, S., 1979. Simulation linking EMG power spectra to recruitment and rate coding. *Comput. Biol. Med.*, 9: 213-221.
- Hagberg, M., 1981. Muscular endurance and surface electromyogram in isometric and dynamic exercise. *J. Appl. Physiol.* 51: 1-7.
- Hagberg, M. and Ericson, B.E., 1982. Myoelectric power spectrum dependence on muscular contraction level of elbow flexors. *Eur. J. Appl. Physiol.*, 48: 147-156.
- Inman, V.T., Ralston, H.J., Saunders, J.B. de C.M., Feinstein, B. and Wright, E.W., 1952. Relation of human electromyogram to muscular tension. *Electroencephalogr. Clin. Neurophysiol.*, 4: 187-194.
- Kadefors, R., Kaiser, E. and Petersen, I., 1968. Dynamic spectrum analysis of myo-potentials and with special reference to muscle fatigue. *Electromyography*, 8: 39-73.
- Kaiser, E. and Petersen, I., 1963. Frequency analysis of muscle action potentials during tetanic contraction. *Electromyography*, 3: 5-17.
- Kogi, K. and Hakamada, T., 1962. Slowing of surface electromyogram and muscle strength in muscle fatigue. *Institute for Science of Labor, Report*, pp. 27-41.
- Komi, P.V. and Buskirk, E.R., 1970. Reproducibility of electromyographic measurements with inserted wire electrodes and surface electrodes. *Electromyography*, 10: 357-367.
- Komi, P.V. and Tesch, P., 1979. EMG frequency spectrum, muscle structure, and fatigue during dynamic contractions in man. *Eur. J. Appl. Physiol.*, 42: 41-50.
- Kondraske, G.V., Deivanayagam, S., Carmichael, T., Mayer, T.G. and Mooney, V., 1987. Myoelectric spectral analysis and strategy for quantifying trunk musculature fatigue. *Arch. Phys. Med.*, 68: 103-110.
- Kroemer, K.H.E., 1974. Designing for muscular strength of various populations. *Wright-Patterson AFB, OH, Aerospace Medical Research Laboratory, AMRL-TR-72-46*.
- Lind, A.R. and Petrofsky, J.S., 1979. Amplitude of the surface electromyogram during fatiguing isometric contractions. *Muscle & Nerve*, 2: 257-264.
- Lindstrom, L., Kadefors, R. and Petersen, I., 1977. An electromyographic index for localized muscle fatigue. *J. Appl. Physiol.*, 43: 750-754.
- Lloyd, A.J., 1971. Surface electromyography during sustaining isometric contractions. *J. Appl. Physiol.*, 30: 713-719.
- Merletti, R., Sabbahi, M.A. and De Luca, C.J., 1984. Median frequency of the myoelectric signal: effects of muscle ischemia and cooling. *Eur. J. Appl. Physiol.*, 52: 258-265.
- Milner-Brown, H.S. and Stein, R.B., 1975. The relation between the surface electromyogram and muscle force. *J. Physiol.*, 246: 549-569.
- Moritani, T., Muro, M. and Nagata, A., 1986. Intramuscular and surface electromyogram changes during muscle fatigue. *J. Appl. Physiol.*, 60: 1179-1185.

- Naeije, M. and Zorn, H., 1981. Changes in the power spectrum of the surface electromyogram of the human masseter muscle due to local muscular fatigue. *Arch. Oral Biol.*, 26: 409–412.
- Naeije, M. and Zorn, H., 1982. Relation between EMG power spectrum shifts and muscle fiber action potential conduction velocity changes during local muscle fatigue in man. *Eur. J. Appl. Physiol.*, 50: 23–33.
- Nanthavanij, S., 1987. ONE FOR ALL or ALL FOR ONE: the role of individual quadriceps muscle components in isometric knee extension: In: S. Asfour (Ed.), *Trends in Ergonomics/Human Factors IV (Proc. Annual International Industrial Ergonomics and Safety Conference)*. Elsevier, Amsterdam, pp. 709–716.
- Petrofsky, J.S., 1980. Computer analysis of the surface EMG during isometric exercise. *Comput. Biol. Med.*, 10: 83–95.
- Petrofsky, J.S. and Lind, A.R., 1980. Frequency analysis of the surface electromyogram during sustained isometric contractions. *Eur. J. Appl. Physiol.*, 43: 173–182.
- Piper, H., 1912. *Elektrophysiologie Menschlicher Muskeln*. Springer, Berlin, Heidelberg, New York, 126 pp.
- Pollak, V., 1980. Surface EMG and muscle forces at low force levels. *Amer. J. Phys. Med.*, 59: 126–141.
- Sadoyama, T. and Miyano, H., 1981. Frequency analysis of surface EMG to evaluation of muscle fatigue. *Eur. J. Appl. Physiol.*, 47: 239–246.
- Sadoyama, T., Masuda, T. and Miyano, H., 1983. Relationships between muscle fiber conduction velocity and frequency parameters of surface EMG during sustained contraction. *Eur. J. Appl. Physiol.*, 51: 247–256.
- Thorstensson, A. and Karlsson, J., 1976. Fatiguability and fiber composition of human skeletal muscle. *Acta Physiol. Scand.*, 98: 318–332.
- Van Boxtel, A. and Schomaker, L.R.B., 1984. Influence of motor unit firing statistics on the median frequency of the EMG power spectrum. *Eur. J. Appl. Physiol.*, 52: 207–213.
- Viitasalo, J.T. and Komi, P.V., 1975. Signal characteristics of EMG with special reference to reproducibility of measurements. *Acta Physiol. Scand.*, 93: 531–539.
- Viitasalo, J.T. and Komi, P.V., 1977. Signal characteristics of EMG during fatigue. *Eur. J. Appl. Physiol.*, 37: 111–121.
- Viitasalo, J.T. and Komi, P.V., 1978a. Isometric endurance, EMG power spectrum, and fiber composition in human quadriceps muscle. *Biomechanics*, VI-A: 244–250.
- Viitasalo, J.T. and Komi, P.V., 1978b. Interrelationships of EMG signal characteristics at different levels of muscle tension and during fatigue. *Electromyogr. Clin. Neurophysiol.*, 18: 167–178.